



Audio Plugin - User Manual & Technical
Documentation

BM11M-

LOW PASS FILTER

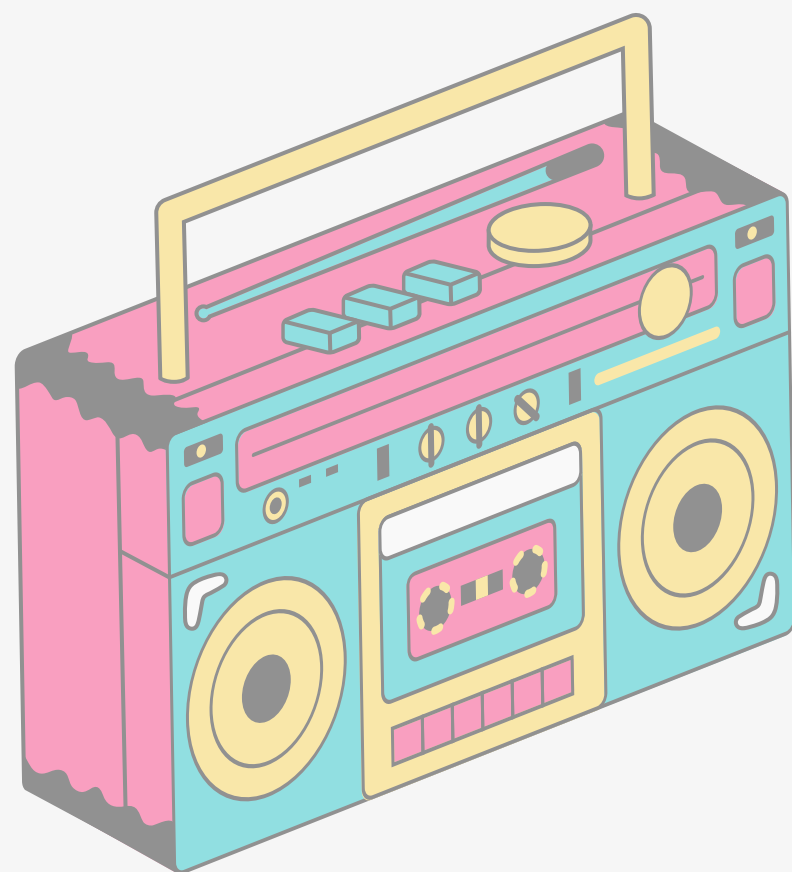
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DAVID
SIGNALS

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Why BM11MLPF? 📝

- **Dynamic Filtering:** Envelope-controlled cutoff modulation
- **2/4-Pole Smoothness:** 12dB & 24dB/octave filter slope
- **Dual Control:** Manual and automatic filtering in one
- **Creative Flexibility:** Perfect for drums, synths, and sound design

INTRODUCTION

Welcome to BM11M Low Pass Filter, a dynamic, envelope-controlled low-pass filter designed for expressive sound shaping and rhythmic filtering effects!. This unique plugin combines a classic 4-pole filter with envelope follower capabilities, allowing your audio's dynamics to automatically control the filter cutoff for musical, responsive effects.





FEATURES OVERVIEW

1

▼ 2 and 4 Pole Low-Pass Filter

- 12dB & 24dB/octave slope for smooth attenuation
- Resonance control for emphasis at cutoff frequency
- Musical filter character

2

📈 Envelope Follower System

- Audio dynamics control filter cutoff
- Adjustable envelope speed (FAST/SLOW)
- Envelope bypass option

3

🔥 Drive

- Pre-filter drive for harmonic excitement
- Analog-style warmth and grit
- Enhances envelope tracking

4

✂️ Dry/Wet Mix Control

- Parallel processing capability
- Blend filtered and dry signals
- Create subtle or dramatic effects





FEATURES OVERVIEW

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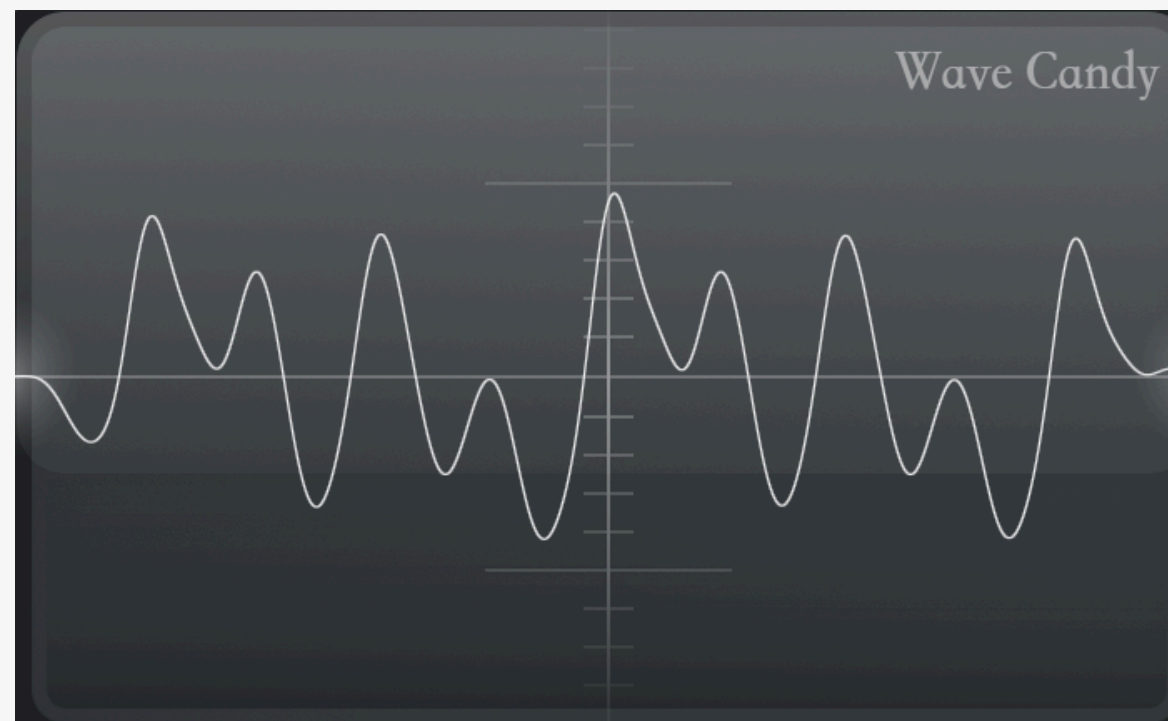
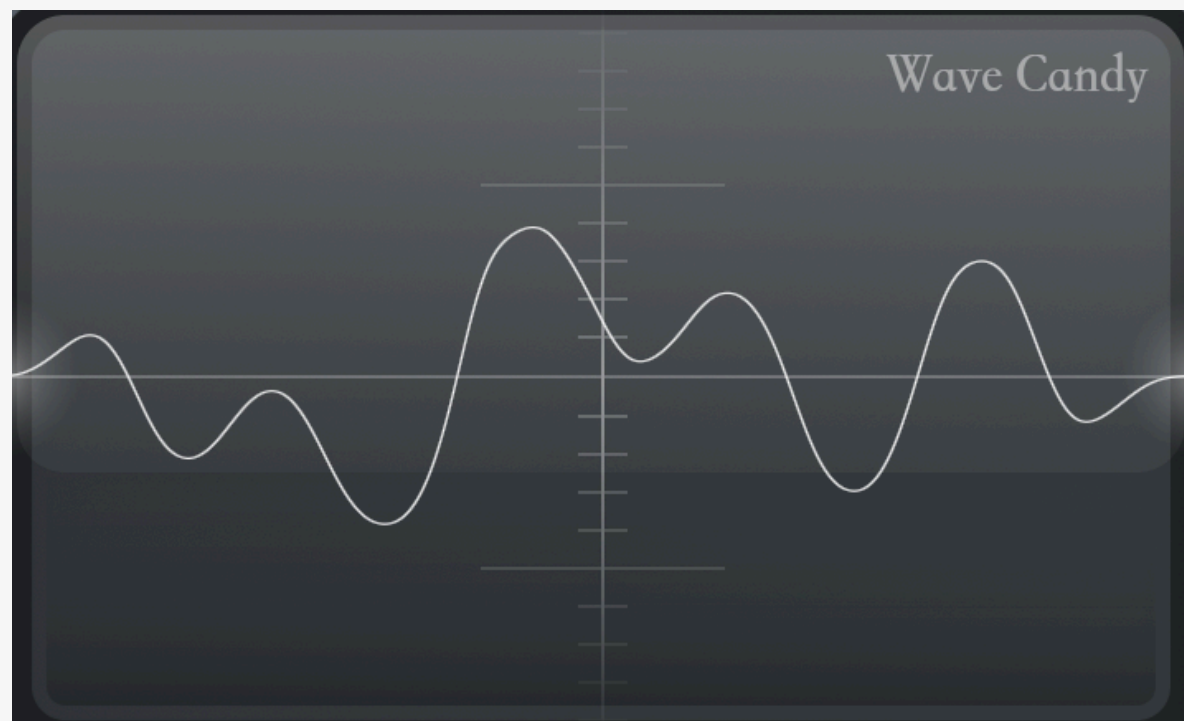


FEATURES OVERVIEW



Envelope Follower System

- Audio dynamics control filter cutoff
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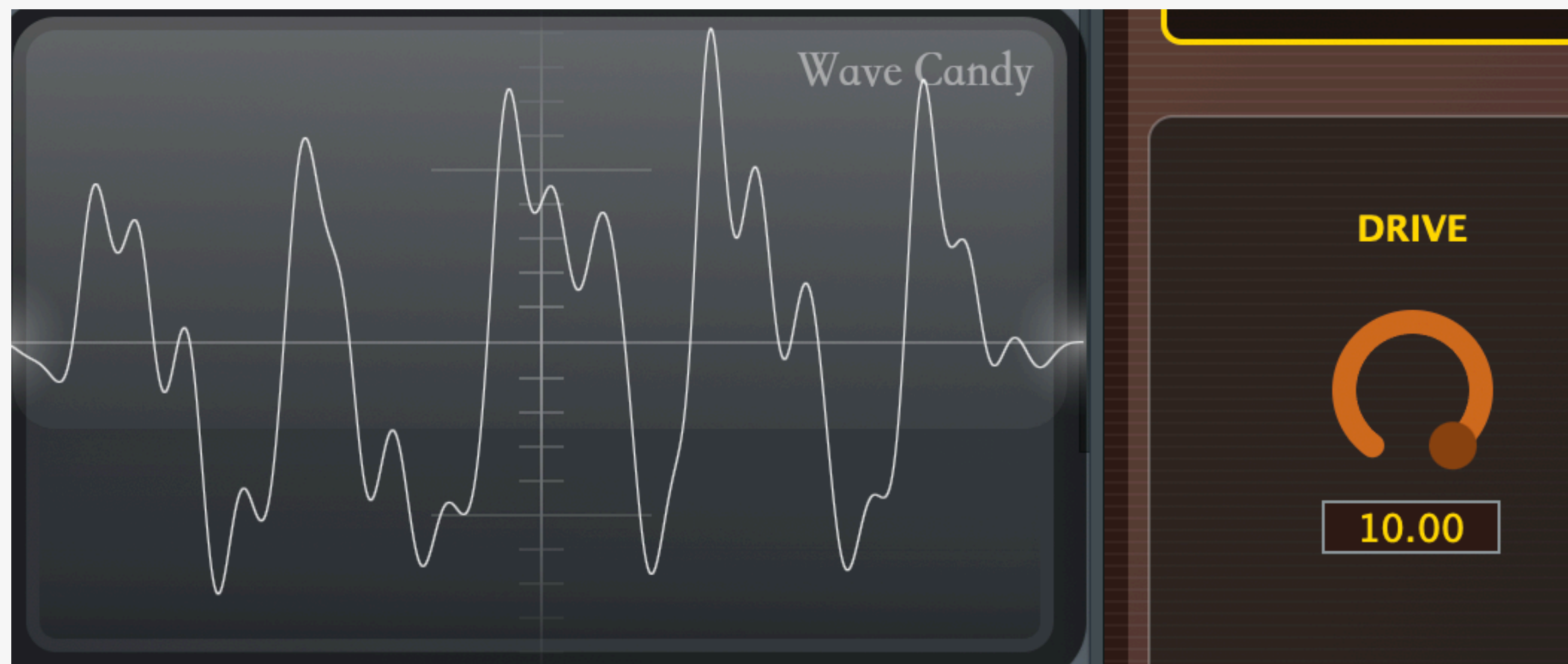




FEATURES OVERVIEW

Drive

- Pre-filter drive for harmonic excitement
- Analog-style warmth and grit
- Enhances envelope tracking



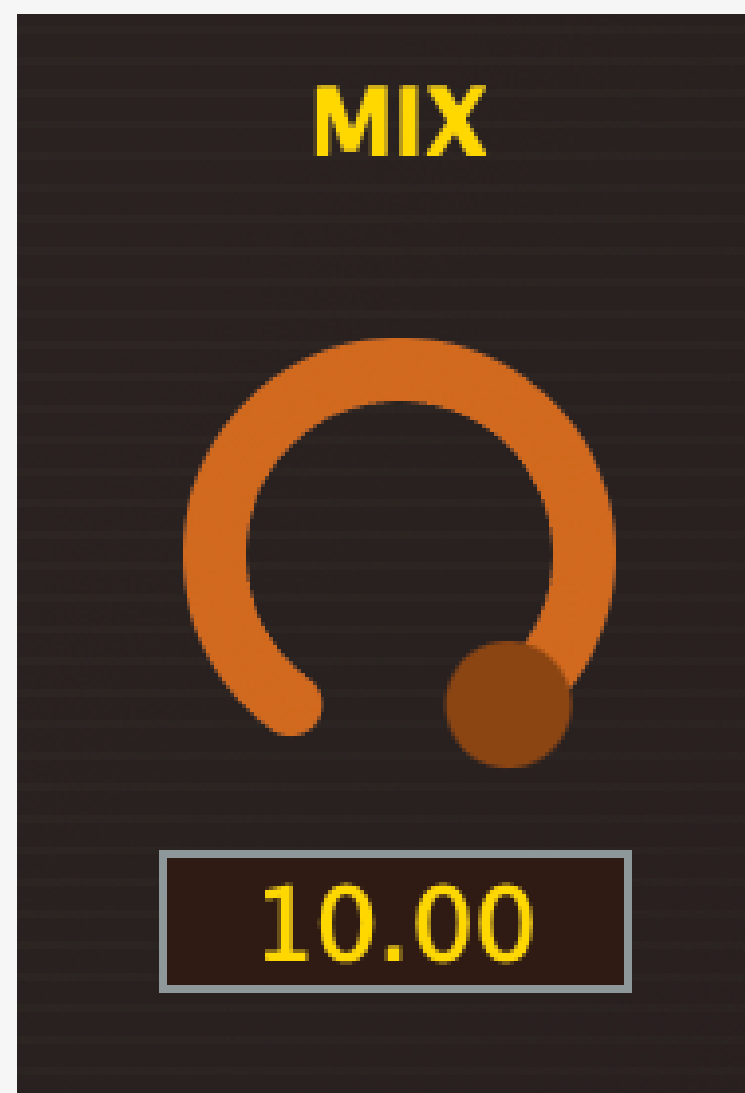


FEATURES OVERVIEW



Dry/Wet Mix Control

- Parallel processing capability
- Blend filtered and dry signals
- Create subtle or dramatic effects



INSTALLATION GUIDE

STEP 1: DOWNLOAD

Download the latest version from the GitHub repository: github.com/DavidSignals/BM11MLPF

STEP 3: RESCAN PLUGINS

- Open your DAW
- Rescan plugin directory
- Locate "BM11MLPF" in your plugin list

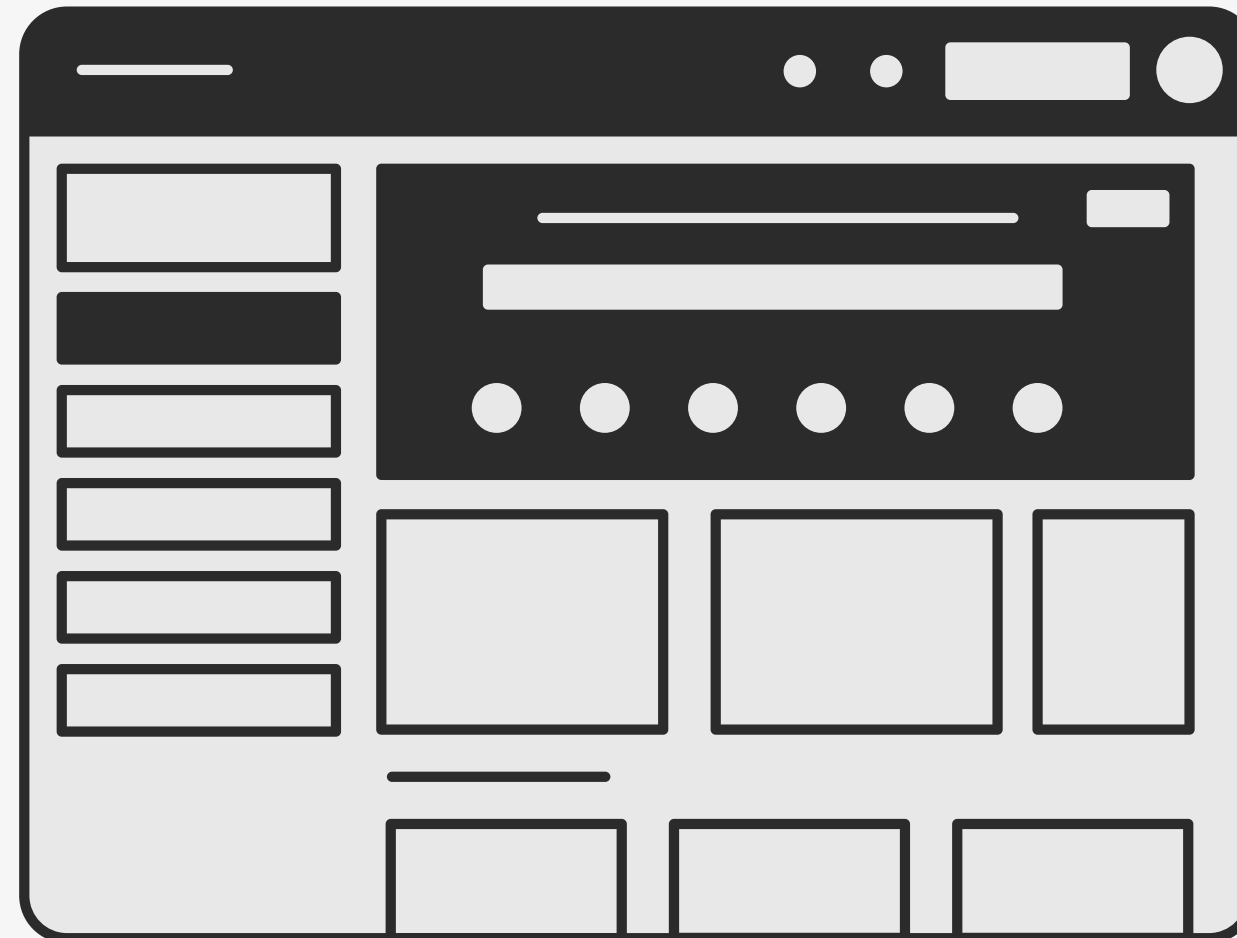
STEP 2: INSTALL PLUGIN FILES

Move the plugin files to the appropriate directories.

- macOS: /Library/Audio/Plug-Ins/VST3 or /Components
- Windows: C:\Program Files\Common Files\VST3

STEP 4: ACTIVATE

Load the plugin on any audio track or bus



System Requirements

- Operating System: macOS 11.0 or later
- Architecture: Apple Silicon & Intel
- DAW Compatibility: All major hosts (Logic Pro, FL Studio, Pro Tools, etc.)
- Plugin Formats: AU, VST3





INTERFACE CONTROL

DRIVE KNOB

- Range: 0.00 to 10.00
- Function: Adds pre-filter saturation and harmonics
- Default: 5.00
- Tip: Increase for more aggressive envelope response

CUTOFF KNOB

- Range: 20 Hz to 20 kHz
- Function: Base cutoff frequency for the filter
- Default: 1000 Hz range
- Tip: Sets starting point before envelope modulation

FILTER SELECTOR

- Option: 4-POLE ✓ (24dB/octave)
- Option 2: 2-POLE ✓ (12dB/octave)
- Function: Selects filter slope steepness
- Note: Currently fixed as 4-pole design

RESONANCE KNOB

- Range: 0.00 to 10.00
- Function: Controls filter resonance/emphasis
- Default: 5.00
- Tip: Higher values create more pronounced filter character

AMOUNT KNOB

- Range: 0.00 to 10.00
- Function: Controls envelope modulation intensity
- Default: 5.00
- Tip: How much the envelope affects cutoff frequency

ENVELOPE SELECTOR

- Options: FAST ✓ / SLOW
- Function: Sets envelope follower response time
- FAST: Quick response for transients (drums, percussion)
- SLOW: Smooth response for sustained sounds (pads, vocals)

MIX KNOB

- Range: 0.00 to 10.00 (0% to 100%)
- Function: Dry/wet blend control
- Default: 5.00 (50% wet)
- Tip: Full wet for full effect, lower for parallel processing

BYPASS SWITCH

- Option: BYPASS ✓ / LEVEL
- Function: Toggles envelope follower on/off
- On: Envelope controls cutoff dynamically
- Off: Static filter operation



PRACTICAL APPLICATIONS



DRUM PROCESSING

- Dynamic filtering on drum buses
- Kick-triggered filter sweeps
- Rhythmic pumping effects
- Hi-hat and percussion excitement

SYNTH & BASS ENHANCEMENT

- Velocity-sensitive filter response
- Expressive playing dynamics
- Auto-wah effects for synths
- Dynamic bass filtering

VOCAL EFFECTS

- Vocal-driven filter movements
- De-essing alternative
- Dynamic vocal presence control
- Talk-box style effects

SOUND DESIGN

- Creating evolving textures
- Ambient pad movement
- Gated filter effects
- Rhythmic pattern generation



TECHNICAL SPECIFICATIONS



AUDIO PROCESSING

- Sample Rates: 44.1kHz to 192kHz
- Bit Depth: 32-bit floating point internal processing
- Latency: Zero additional latency
- Channels: 1-in/2-out, 2-in/2-out

FILTER DATA

- **Type:** 4-Pole Low Pass (24dB/octave) & 2-Pole Low Pass (12dB/octave)
- **Resonance:** Adjustable, self-oscillating at high settings
- **Envelope Follower:** Frequency-based dynamics detection

PLUGIN SPECIFICATIONS

- Framework: JUCE 8.0+
- Code: C++17
- UI: Custom OpenGL rendering
- Parameters: AudioProcessorValueTreeState managed



TROUBLESHOOTING

No Filter Movement

- Check ENV BYPASS is off
- Increase AMOUNT control
- Ensure audio signal has dynamics

Filter Too Aggressive

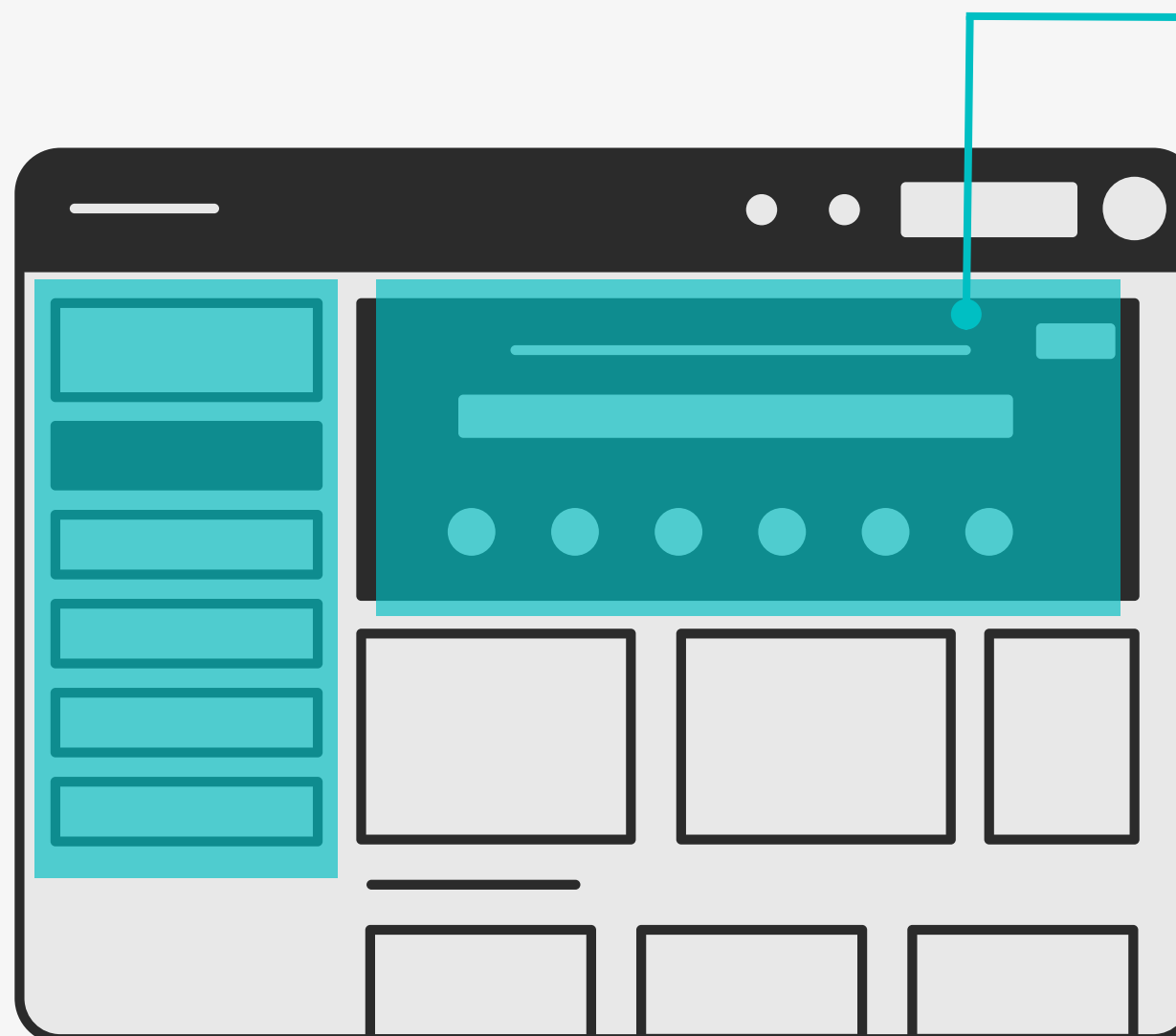
- Reduce RESONANCE
- Lower AMOUNT control
- Adjust ENVELOPE to SLOW setting

Unwanted Oscillation

- Decrease RESONANCE
- Lower DRIVE amount
- Adjust input levels

Plugin Not Detected

- Verify installation path
- Rescan plugins in DAW
- Check plugin format compatibility



CPU Usage High

- BM11MLPF is optimized for low CPU usage
- If issues occur, check other plugins in session

Support Contact

For technical support or bug reports:

- **GitHub Issues:**
github.com/davidsignalsss/bm11mlpf/issues
- **Email:** signalsandstrings@gmail.com
- **Documentation:** Full documentation available on GitHub

David Rábago Félix

Professional Alias: David Signals

Background

Audio plugin developer and mechatronic engineer with a passion for creating tools that enhance creative workflows. Specializing in JUCE framework development and digital signal processing. My hands-on hardware expertise includes designing guitar pedal effects and building synthesizer circuits with integrated circuits and programmable platforms like Raspberry Pi, ESP32, Arduino, Teensy 4.1, and Daisy Seed. Currently designing a multi-effects pedalboard. Guitarist and music lover.

Philosophy

"Great tools should be invisible - they should empower creativity without getting in the way. Mono/Stereo Ears embodies this principle through its intuitive design and reliable performance."

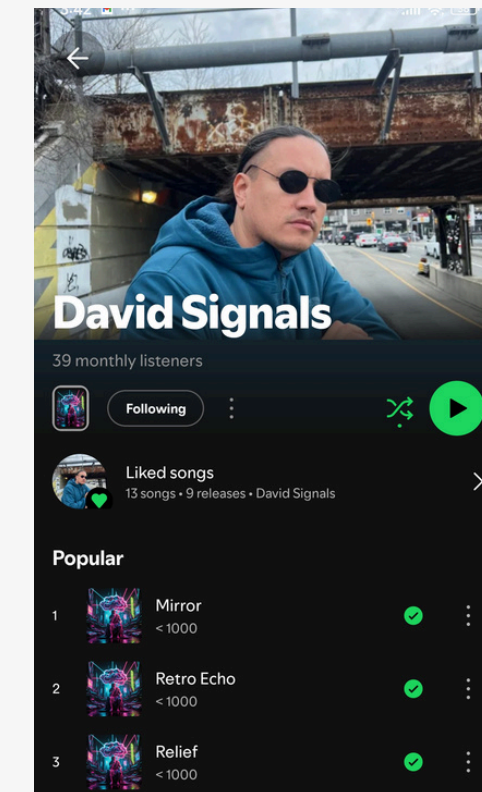
Connect

- GitHub: <https://github.com/DavidSignals>
- LinkedIn: www.linkedin.com/in/davidrabagofelix

Acknowledgments

Special thanks to the JUCE framework team and the audio developer community for their continuous inspiration and support.

ABOUT ME





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